MIL-A-81403(USAF)
AMENDMENT 2
18 February 1986
SUPERSEDING
AMENDMENT 1
30 NOV 1979

MILITARY SPECIFICATION

ALTIMETER-ENCODER AAU-21/A

This amendment forms a part of Military Specification MIL-A-81403(USAF), dated 15 June 1967, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

* 1.1 Add title: Scope.

* Add new paragraph:
"2.1 Government documents."

* Delete existing paragraph number 2.1 and the following source statement and add:
"2.1.1 Specifications, Standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed on the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under SPECIFICATIONS Federal add: "PPP-B-601 Boxes, Wood, Cleated Plywood"; "PPP-B-636 Box, Shipping Fiberboard".

* Under SPECIFICATIONS Military delete "MIL-E-5373"; "MIL-D-1000/1; "MIL-N-18307".

Add boxed paragraph to bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* Under STANDARDS Military add: "MIL-STD-100 Engineering Drawing Practices"; "MIL-STD-129 Marking For Shipment And Storage"; "MIL-STD-810 Environmental Test Methods And Engineering Guidelines".

PAGE 3

3.1 Delete and add:
"3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.2 and 6.2)".

AMSC N/A FSC 6610

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* 3.3.1 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

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3.11 In first line delete "MIL-N-18307" and add "MIL-STD-130".

3.14 In last line delete "MIL-D-1000/1" and add "MIL-STD-100".

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4. Delete all references to "Qualification" and add "First article", except in 4.2.(b).

4.2.(b) Delete "Qualification conformance inspection" and add "Quality conformance inspection".

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* 4.4.2 Delete and add:
"4.4.2 Sampling Plan B sample selection. Three samples shall be selected at random from the first 15 produced on contract".

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* 4.7.2 In line three delete "procedures of MIL-E-5262." and add "test method and procedure of MIL-STD-810."

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* 4.7.2.2 In line two delete "Procedure II" and add "Method 502.1 Procedure I, except that in Step 2 the storage temperature shall be -62°C (-30°F) and the soak period shall be 72 hours. The operating temperature in Step 4 shall be -54°C (-65°F) and the stabilization period shall be 24 hours".

* 4.7.2.3 in line two delete "Procedure I" and add "Method 501.1 Procedure I, except delete Step 4".

* 4.7.2.7 Delete and add:
"4.7.2.7 Vibration failure. The Altimeter-Encoder shall be mounted on the vibration apparatus in its normal operation position. While being operated at one of the asterisked test points in Table II, shall be vibrated with circular motion of .018 to .020 inch diameter in a plane inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly from 5 to 50 Hertz and return once each hour for a three hour period."

* 4.7.2.8 In line two delete "Procedure I" and add "Method 513.2, Procedure II". To the end of the paragraph add: "During this test the Altimeter-Encoder shall be operated at on of the asterisked test points in Table II. There shall be no malfunction of the Altimeter-Encoder or power warning circuits during applied acceleration".

* 4.7.2.9 In line two delete "Procedure V, operational shock only" and add "Method

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516.2-2. amplitude of 15 g's peak acceleration, duration of 11±1 milli-seconds. During this test Alimeter-Encoder shall be installed in its normal operating position, with external lines installed, but not be operational".

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* 4.7.2.13 In line two delete "Procedure I" and add "Method 507.1, Procedure I, except that the chamber temperature of Step 2 shall be 71°C (160°F) and the relative humidity of Step 4 shall be 95 percent".

* 4.7.2.14 In lines one and two delete "Procedure I" and add "Method 508.1, Procedure I".

* 4.7.2.15 In line two delete "Procedure I" and add "Method 509.1, Procedure I".

* 4.7.2.16 In line two delete "Procedure I" and add "Method 510.1, Procedure I, except that the air velocity in Step 1 shall be 300±200 feet per minute. The air temperature in Step 2 shall be 71°C (160°F) and the exposure period shall be six hours. Delete Step 3".

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* Add"
"4.9 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

5. Delete and add:

5 PACKAGING

5.1 Preservation-packaging Preservation-packaging shall be Level A or C as specified by contracting activity.

5.1.1 Level A.

* 5.1.1.1 Cleaning. Altimeter-Encoder shall be cleaned in accordance with process C-1 of MIL-P-116.

* 5.1.1.2 Drying. Immediately after cleaning, the Altimeter-Encoder shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

* 5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each Altimeter-Encoder shall be packaged in quantity unit packs of each in accordance with method IA-8 of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Carton shall be large enough to allow for application of sufficient cushioning material between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the Altimeter-Encoder when completed packs are subjected to the rough handling drop test of MIL-P-116.

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* 5.1.2 Level C. Altimeter-Encoders shall be cleaned, dried and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Altimeter-Encoders packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, styles A or B, class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be uniform shape, size, and minimum tare and cube, consistent with the protection required.

* 5.2.2 Level B. Altimeter-Encoders packaged as specified in 5.1.1 shall be packed in weather resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

* 5.2.3 Level C. Packaged Altimeter-Encoders shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

* 5.3 Marking. In addition to any special marking required by the contract or order, each Altimeter-Encoder package and shipping container shall be marked in accordance with MIL-STD-129."

6.2 (c) Delete "Reliability Qualification" and add "Reliability First Article"

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* 6.4 Delete
The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content previous amendment.

* Certain provisions of this document are subject to international standardization agreement ASCC AIR-STD 10/60C. When amendment, revision, or cancellation of this document is proposed which will modify the international agreement concerned, the preparing activity takes appropriate action through international standardization channels including the Departmental Standardization Offices (DepSo) to change the agreement or make other appropriate accomodations.

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